

Grade 3 Noetic Math: Challenge Practice Set

Time Limit: 45 Minutes | 20 Questions | No Calculators

Part 1: Numbers & Logic Puzzles

1. If $3A + 5A + AA = 167$, what is the value of digit A ?
2. What is the largest 3-digit number that is a multiple of both 7 and 9?
3. Using the digits 0, 4, and 7 exactly once each, how many different 3-digit numbers can you make?
4. Find the sum of all two-digit numbers where the tens digit is twice the ones digit.
5. In a sequence, the first number is 3. Each following number is found by doubling the previous number and adding 1. What is the 8th number?

Part 2: Multi-Step Word Problems

6. Kevin and Sarah have 120 stamps. After Kevin gives 20 to Sarah, Sarah has 10 more than Kevin. How many did Sarah have at first?
7. 2 big dogs weigh the same as 5 small cats. If 3 cats and 1 dog weigh 44 lbs total, what does one big dog weigh?
8. A basket has apples, bananas, and pears. There are 4 more apples than bananas, and twice as many pears as apples. With 28 fruits total, how many are bananas?
9. Sam gives half his marbles to his brother, wins 12 more, then gives 5 to a friend. He has 25 left. How many did he start with?
10. A movie starts at 1:45 PM and is 135 minutes long. What time does it end?

Part 3: Geometry & Measurement

11. A garden is 10m by 6m. A 2m-wide path runs inside the edge. What is the area of the remaining garden space?
12. A large rectangle is made of 3 identical squares side-by-side. If the total perimeter is 64 cm, what is the area of one square?
13. A rectangle has a perimeter of 48 cm. Its length is twice its width. What is its area?
14. A 4x4 grid of small squares forms a large square. If the area of the large square is 64 sq cm, what is the perimeter of one small square?
15. A fence is 40 feet long. Posts are placed every 5 feet, including the ends. How many posts are there?

Part 4: Advanced Logic & Probability

16. There are 5 red, 4 blue, and 3 green pens. If you pick one without looking, what is the probability it is not blue? (Give as a simplified fraction).
17. Three friends share \$9.00 in a ratio of 2:3:4. How much does the person with the most money get?
18. On a 12-hour digital clock, what is the largest possible sum of the digits shown?
19. A book has 100 pages. How many times does the digit "7" appear in the page numbers?
20. Five players scored a total of 15 points. Each scored a different number of points, and everyone scored at least 1. How many points did the top scorer get?